

EHSAN SHARAFIAN

Curriculum Vitae

✉ : ehsan.sharafian@maine.edu

☎ : (207) 631-6979



EDUCATION

PhD:

- Mechanical Engineering
 - **University of Maine**, Orono, ME, US
 - Total GPA: 4/4
 - Thesis: Smart Real-time Gait Rehabilitation System
- Sep 2023 - Present

Master of Science:

- Mechanical Engineering
 - Robotics and Control Systems
 - **Amirkabir University of Technology (Tehran Polytechnic)**
 - Total GPA: 3.65 / 20
 - Thesis: Trajectory optimization of a robot arm in the joint and Cartesian spaces
- Sep 2017 – Feb 2020

Bachelor of Science:

- Mechanical Engineering
 - **University of Tehran**
 - Total GPA: 14.30 / 20
- Sep 2012 - Jul 2017

PUBLICATIONS

- **Sharafian Moghaddam, Ehsan**, and Babak Hejrati. "Real-Time Foot Height Estimation and Activity Classification Using a Foot-Mounted IMU Implemented on a Smartphone." *Sensors* (2026). May 2026
- **Sharafian, M. Ehsan**, Colby Ellis, and Babak Hejrati. "Real-Time Activity Recognition Using Minimal Biomechanical Features: A Lightweight IMU-Based Classifier for Older Adults." *IEEE Access* (2025). Sep 2025
- **Sharafian, M. Ehsan**, et al. "The effects of real-time haptic feedback on gait and cognitive load in older adults." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2025). Jun 2025
- Noghani, M. A., **Sharafian, E.**, Sidaway, B., & Hejrati, B. (2025). Increasing thigh extension with haptic feedback affects leg coordination in young and older adult walkers. *Journal of Biomechanics*, 112525. Jan 2025

- **Sharafian E.**, Ghassabzadeh M., Taghvaeipour A., "Trajectory optimization of a spot-welding robot in the joint and Cartesian spaces," International Journal of Robotics and Automation (IRJA), DOI: 10.2316/J.2022.206-0701 Jan 2023
- **Sharafian, Ehsan**, Afshin Taghvaeipour, and Maryam Ghassabzadeh. "Revisiting screw theory-based approaches in the constraint wrench analysis of robotic systems." Robotica 40.5 (2022): 1406-1430. Sep 2021

Journal Reviewers

- Human Science Reports 1 paper
- IEEE Journal of Biomedical and Health Informatics 1 paper
- International Journal of Robotics and Automation 1 paper

ACADEMIC EXPERIENCE

- Laboratory Research Assistant
- New Technology Research Center (NTRC) of Amirkabir University of Technology
 - Assist in conducting undergraduate research projects
 - The provision of laboratory services for graduate students Nov 2020 - Nov 2022
 - Maintaining the laboratory equipment and following up on the repairments
 - Cooperation in holding the RSI 9th ICROM Conference (International Conference on Robotics and Mechatronics)

WORK EXPERIENCE

- Sina Robotics and Medical Innovators Co
- Fabrication and development in the field of Medical and Surgical Robotics
- *Products:* Surgical Robots, Surgeon-Assistant Robots, Assistive and Therapeutic Rehabilitation robots, and advanced medical simulators for training purposes Jan 2023 – Sep 2023
- *My Role:* Junior Mechanical and Control Systems Engineer
 - ✓ Developing Control systems of the robots for Laparoscopic surgery
 - ✓ Designing interfaces software of the robots for human interaction
 - ✓ Training surgeons doctors and medical students on how to use the robot

TEACHING EXPERIENCES

- 1 Semester as Teaching Assistant for MEE 370 "Manufacturing design" Sep 2023 – Dec 2023
- 2 Semesters as Teaching Assistant for "Dynamics 2", undergraduate course Jan 2020 - Jun 2020 & Jan 2021 - Jun 2021
- 1 Semester teaching Assistant for "Dynamics of Machines", undergraduate course Jan 2020 - Jun 2020
- 4 Years teaching courses on the university entrance exam to private students 2013 - 2017

LANGUAGE PROFICIENCY and GRE TEST

TOEFL iBT Score:

- Total: **98** (R: 25, L: 27, S: 24, W:22)
- Test date: November 28, 2021

GRE Score:

- W: **4**, Q:**169**, V: 144
- Test date: July 5, 2022

CERTIFICATES

- Prompt Engineering for ChatGPT [↗](#) April 2023
- Convolutional Neural Networks: Coursera: DeepLearning.AI (Andrew Ng) [↗](#) Aug 2021
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization: Coursera: DeepLearning.AI (Andrew Ng) [↗](#) Aug 2021
- Neural Network and Deep Learning: Coursera: DeepLearning.AI (Andrew Ng) [↗](#) Jul 2022
- Introduction to Machine Learning: Coursera: Duke University [↗](#) May 2022

SKILLS

Relevant Software

- Matlab
- Solidwork
- SPSS
- Android Studio
- Adams
-

Programming Languages

- Python
- C/C++
- C#
- Java

HONORS & AWARDS

- Graduate Student Associations (GSG) Awards for conference proposal March 2025
- Receiving the Flagship fellowship Award Sep 2024 – Sep 2025

- Ranked 316th among more than 400,000 participants in the national university entrance exam for Bachelor's degree Jul 2012
- Ranked 216th among more than 20,000 participants in the national university entrance exam for Master's degree June 2017
- Waiver of full tuition fees for Bachelor's degree Sep 2012 - July 2017
- Waiver of full tuition fees for Master's degree Sep 2017 - Feb 2020
- 3rd place in group teams at the RoboCup Junior in Tehran Dec 2010

REFERENCES

- Babak Hejrati
Associate Professor,
Department of Mechanical Engineering
University of Maine, Orono, ME, US
Email: Babak.hejrati@maine.edu
Phone: (801) 739 2052

Philippe Malcolm
Associate Professor,
Department of Biomechanics
University of Nebraska, Omaha

- Email: pmalcolm@unomaha.edu
- Phone: (617) 487-1148

- Ben Sidaway
Geneva Allen '49 Professor in Health Sciences, College of Health and
Pharmacy, School of Physical Therapy
Husson University, Bangor, ME
Email: sidawayb@husson.edu
Phone: (207) 973 1099

- Dr. Mahnaz Shamshirsaz (NTRC Dean)
Professor,
Head of New Technology Research Center (NTRC) of Amirkabir University of
Technology (Tehran Polytechnic)
Email: shamshir@aut.ac.ir
- Phone: +98 (21) 645-45191